TECHNICAL BENCHMARK GUIDE

SpeedCloud Supreme

How to read the June 2026 performance report

A practical interpretation of the supplied benchmark report: the tools used, the reported measurements, what each metric means, where the results may help, and how to validate them against a real workload.

NxtGen | SpeedCloud Supreme

Prepared from SpeedCloud Supreme Performance Benchmark - June 2026 V5

01 | HOW TO READ THE REPORT

A benchmark is evidence about a test, not a universal promise

The supplied report evaluates four infrastructure pillars using Geekbench 6 for compute, FIO for storage, iPerf for network throughput, and Sysbench for memory. The results are useful signals for platform screening, but a buying decision should also use the workload, configuration, test duration, concurrency, protection overhead, and service design that apply to the customer.

Pillar	Tool in report	What it measures	Example workload connection
Compute	Geekbench 6	Single-core and multi-core processor behavior across general and compute-intensive tasks.	Application responsiveness, VM density, build jobs, analytics, and transaction processing.
Storage	FIO	IOPS, bandwidth, average latency, and p99 latency for random, mixed, and sequential access.	Databases, VM datastores, backup streams, file services, and large data movement.
Network	iPerf	Sustained sender and receiver throughput across repeated tests.	Replication, application tiers, backup traffic, data pipelines, and east-west traffic.
Memory	Sysbench	Data transferred, transfer rate, and maximum latency for memory operations.	Caching, databases, VM efficiency, analytics, and real-time processing.

What the report does establish

- The tested platform produced strong processor, storage, network, and memory results in the stated benchmark runs.
- The storage results include both average and p99 latency for the 4K tests, which helps assess consistency under load.
- The network runs show approximately 30 to 31 Gbits/sec with closely matching sender and receiver rates.

What still needs confirmation

- The report does not provide a full test configuration, sample size, run duration, queue depth, concurrency, host model, or virtualization overhead.
- It does not compare against a named baseline, customer environment, or equivalent protected production configuration.
- It does not by itself prove application response time, availability, backup recovery, SLA performance, or cost efficiency.

02 | COMPUTE PERFORMANCE

Geekbench 6 indicates processor capacity, not an application SLA

The report records a single-core score of 2017 and a multi-core score of 3947. Single-core behavior is relevant when a critical thread cannot be parallelized; multi-core behavior is relevant when the platform can distribute work across concurrent threads or VMs.

Metric	Reported result	Technical meaning	Where it can help
Single-core score	2017	A relative indicator of one-thread execution capability in the Geekbench 6 run.	Interactive application paths, licensing-limited software, serial jobs, and transaction steps.
Multi-core score	3947	A relative indicator of aggregate processor behavior when work can use several cores.	Concurrent VMs, parallel builds, analytics workers, batch processing, and multi-user services.

How to apply the result

- For a VM estate, compare the expected vCPU-to-core ratio, CPU ready time, contention, and noisy-neighbor controls; a host score is not the same as a VM guarantee.
- For databases, test the real query mix and concurrency because CPU performance may be limited by storage latency, locks, memory, or licensing.
- For application servers, measure requests per second, p95 or p99 response time, and saturation behavior rather than using Geekbench as the only acceptance criterion.

Example: CPU-oriented proof of concept

Run a representative API or transaction workload at normal and peak concurrency. Record CPU utilization, throttling, response-time percentiles, error rate, and throughput. Repeat with the same software version, data volume, and concurrency on the proposed target. The benchmark report can support platform screening; the application test decides fit.

Important reading note

The report presents the Geekbench results as evidence of responsiveness, concurrent workload handling, and virtualization suitability. Treat those as technical interpretations, then validate them against the customer workload and the agreed resource policy.

03 | STORAGE PERFORMANCE: 4K WORKLOADS

IOPS and latency describe transactional behavior

The 4K FIO results are most relevant to small-block random and mixed access. These patterns commonly appear in databases, VM boot and OS activity, metadata-heavy applications, and transactional systems. p99 latency is especially important because users experience slow outliers, not just the average operation.

Test	IOPS	Bandwidth	Avg latency	p99 latency
Random read	94.1k	368 MiB/s	10.87 ms	20 ms
Random write	61.3k	239 MiB/s	16.7 ms	40 ms
Mixed read	38.6k	151 MiB/s	12.78 ms	32 ms
Mixed write	38.6k	151 MiB/s	13.74 ms	34 ms
Sequential read	249k	973 MiB/s	4.11 ms	9.37 ms
Sequential write	74.3k	290 MiB/s	13.77 ms	33 ms

What these numbers mean

- 94.1k random read IOPS suggests the tested storage can service many small reads in parallel; it is a useful signal for read-heavy transactional or VM workloads.
- 61.3k random write IOPS and a 40 ms p99 show write capability with higher tail latency than reads; write-heavy applications should test commit and log behavior directly.
- The mixed test reports balanced 38.6k read and 38.6k write IOPS, with similar average latency, which is relevant to applications that read and write at the same time.
- The sequential read result of 973 MiB/s is relevant to ordered data movement such as backups, scans, and large file reads; it should not be substituted for random I/O behavior.

Example: database and VM validation

Run a database workload with data, log, and temporary areas separated as they would be in production. For a VM estate, measure boot storms, patch windows, snapshot behavior, backup jobs, and recovery. Capture average and p99 latency, queue depth, throughput, IOPS, CPU overhead, and failure behavior.

04 | STORAGE PERFORMANCE: 64K THROUGHPUT

Large-block results describe sustained data movement

The report also uses a 64K block size. Larger blocks require fewer I/O operations to move the same amount of data, so the emphasis shifts from small-operation responsiveness toward throughput for bulk transfers, backups, analytics, media, and large datasets.

Test	IOPS	Bandwidth	Avg latency
Random read	23.7k	5919 MiB/s	10.79 ms
Random write	6990	1748 MiB/s	36.61 ms
Mixed read	4924	1231 MiB/s	26.95 ms
Mixed write	4964	1241 MiB/s	24.81 ms
Sequential read	25.8k	6439 MiB/s	9.92 ms
Sequential write	9994	2499 MiB/s	25.61 ms

How to interpret the 64K results

- The 5919 MiB/s random-read and 6439 MiB/s sequential-read results indicate strong large-block read throughput in the tested configuration.
- The 1748 MiB/s random-write and 2499 MiB/s sequential-write results show that sustained writes are measurable and materially more demanding than reads.
- Mixed read/write throughput is close between the two directions, which is useful evidence for balanced bulk traffic, but the application must still test its own read/write ratio.
- Because the report does not show the 64K p99 values in the summary table, use the average latency only as a directional indicator and ask for tail-latency data if the workload is latency-sensitive.

Example: backup and analytics planning

Estimate the data set size, backup window, retention policy, and concurrency. Compare the required transfer rate with the measured bandwidth after encryption, deduplication, compression, network, and protection overhead. For analytics, test the same data format and query plan because scan throughput can change with compression and parallelism.

05 | NETWORK PERFORMANCE

iPerf shows high sustained throughput; latency and loss need separate tests

The iPerf section reports three runs at 30.2, 30.7, and 31 Gbits/sec, with sender and receiver values matching in each run. The transfers were 35.1 GB, 35.7 GB, and 36.1 GB respectively. This is strong evidence of high-throughput network capability in the tested path.

Run	Data transferred	Sender	Receiver	Technical reading
Test 1	35.1 GB	30.2 Gbits/sec	30.2 Gbits/sec	Sustained high-rate transfer with matching endpoints.
Test 2	35.7 GB	30.7 Gbits/sec	30.7 Gbits/sec	Throughput remains stable across a repeated run.
Test 3	36.1 GB	31.0 Gbits/sec	31.0 Gbits/sec	The path sustains approximately 31 Gbits/sec in the reported run.

Where this matters

- Backup, replication, and large data transfers can complete faster when the storage and network paths are both capable of sustaining the required rate.
- Application tiers can benefit from high east-west bandwidth, but response time also depends on application protocol, serialization, storage, and CPU.
- Matching sender and receiver throughput suggests the test did not show a major throughput mismatch; it is not the same as proving zero packet loss or low latency.

Network questions to close

- Ask for round-trip latency, jitter, packet loss, MTU, oversubscription, and failure behavior between the actual application zones.
- Repeat the test through firewalls, load balancers, encryption, backup software, or replication products if those components are in the production path.
- Confirm whether the result is host-to-host, VM-to-VM, same-rack, cross-zone, or cross-site; topology changes the meaning of the number.

Example: replication acceptance test

Measure the time to replicate a known data set while the application is active. Record throughput, lag, CPU, storage impact, packet loss, and recovery-point behavior. Acceptance should use the business RPO and recovery window, not only the iPerf headline rate.

06 | MEMORY PERFORMANCE

Sysbench shows memory movement and maximum latency

The memory section reports 76,597.22 MiB of data transferred, a 7,658.72 MiB/sec transfer rate, and maximum latency of 0.45 ms. These values indicate strong memory activity in the benchmark run, but memory capacity, NUMA placement, cache behavior, and application-level performance still need workload testing.

Metric	Reported result	Technical meaning	Example use
Data transferred	76,597.22 MiB	The volume of memory data processed during the run.	Large working sets, caching, analytics, and VM memory activity.
Transfer rate	7,658.72 MiB/sec	The reported rate of moving data through the memory subsystem.	Memory-intensive application phases and concurrent VM execution.
Maximum latency	0.45 ms	The highest latency reported by this benchmark run, as presented in the source report.	Latency-sensitive processing, caching, and real-time application paths.

How to apply the result

- Check whether the workload is memory-capacity constrained before focusing on transfer rate; swapping or memory pressure can dominate application behavior.
- For virtualized workloads, validate NUMA alignment, vCPU and memory overcommit, ballooning, page sharing, and contention policies.
- For databases and analytics, measure cache hit rate, query time, garbage collection, spill-to-disk, and concurrent-user behavior.
- Ask for the Sysbench command, thread count, memory size, and test duration so the result can be repeated under controlled conditions.

Example: memory-sensitive application test

Use a representative data set and concurrency. Record working-set size, cache hit rate, CPU utilization, memory bandwidth, page faults, latency percentiles, and throughput. Increase concurrency until the workload reaches its operating limit, then compare the saturation point and recovery behavior.

07 | MAPPING METRICS TO WORKLOADS

Use the right benchmark for the question you are asking

A platform can look strong in one metric and still be a poor fit for a specific application. Match each workload to the measurements that control its user experience, then test the complete stack with realistic data and protection settings.

Workload	Most relevant signals	Example validation
Transactional database	Random read/write IOPS, p99 storage latency, CPU single-core behavior, memory latency, and log throughput.	Run representative queries, commits, checkpoints, backup, and failover with production-like concurrency.
VM estate	Mixed storage behavior, CPU multi-core score, memory capacity and contention, boot storms, and network throughput.	Test simultaneous boot, patch, snapshot, backup, and application activity across a representative VM wave.
Backup and replication	Sequential read/write throughput, network throughput, encryption overhead, and recovery performance.	Measure backup window, replication lag, restore time, and application impact at the required retention and concurrency.
Analytics and batch	Multi-core compute, memory throughput, sequential storage, network movement, and data pipeline concurrency.	Run the real job or query plan with the same data volume and measure completion time, spill, and cost per run.
Modern application services	Single-core response, network latency and loss, storage latency, memory behavior, and autoscaling response.	Measure p95/p99 API latency, requests per second, error rate, scaling time, and dependency behavior.

Example: why one headline number is not enough

A backup system may benefit from the 6,439 MiB/s sequential-read result, but it may still miss its recovery window if encryption, deduplication, network distance, or target storage becomes the bottleneck. A database may benefit from 94.1k random-read IOPS, but its user response time can still be limited by p99 write latency, locking, CPU serialization, or application code. The end-to-end test should follow the business path.

Recommended evidence pack

- Benchmark command lines, configuration, host and VM details, block sizes, queue depth, thread count, duration, and repeat count.
- Raw output and time-series graphs, not only a selected headline result.
- Workload baseline, target test, protection overhead, failure test, and a written interpretation of the differences.
- A mapping from measured result to business objective such as response time, completion window, RPO, RTO, or cost per transaction.

08 | PROOF-OF-CONCEPT AND ACCEPTANCE PLAN

Turn the report into a repeatable technical decision

The benchmark report is a strong starting point for platform qualification. The next step is a controlled proof of concept that combines compute, storage, network, memory, operations, and protection so the customer can make a workload-specific decision.

Phase	Activities	Exit evidence
1. Baseline	Capture current workload size, utilization, response time, IOPS, latency, throughput, memory pressure, backup, and recovery behavior.	Signed baseline with workload owners, data volume, concurrency, and business targets.
2. Reproduce	Repeat the reported tools with documented settings; confirm that the supplied measurements can be reproduced or explained.	Versioned commands, raw results, configuration, and repeatability notes.
3. Workload test	Run representative applications, databases, VM waves, backup, network flows, and memory-intensive jobs.	Application metrics, saturation behavior, error rate, and operator observations.
4. Failure and protection	Test backup, restore, replication, network interruption, storage pressure, and planned maintenance.	Restore evidence, RPO/RTO result, failure impact, recovery steps, and rollback decision.
5. Decision	Compare results with thresholds, cost assumptions, support boundaries, and migration effort.	Go, revise, or stop decision with an agreed action list and commercial assumptions.

Acceptance criteria examples

- Application p95 response time stays below the agreed threshold at peak concurrency, with no material increase in error rate.
- Database p99 read and write latency, transaction rate, and checkpoint behavior meet the workload target.
- Backup completes within the window and restore meets the required RTO with the required data consistency.
- Replication lag remains within the RPO during normal load and during the agreed peak scenario.
- The three-year cost model includes compute, storage, network, backup, DR, security, support, migration, growth, and exit assumptions.

Questions for the supplier

- What exact hardware, software versions, virtualization layer, storage layout, and network topology produced the report?
- Which results are dedicated-resource measurements and which may be affected by shared capacity or oversubscription?
- What performance, availability, backup, DR, and support commitments are contractual rather than indicative?
- How are noisy neighbors, maintenance, capacity expansion, and performance degradation handled operationally?

09 | DECISION SUMMARY

Strong infrastructure signals, with workload validation still required

The supplied report presents SpeedCloud Supreme as a high-performance private cloud platform. Its most useful evidence is the combination of strong processor results, high 4K and 64K storage performance, approximately 30 to 31 Gbits/sec network throughput, and 7.6 GiB/sec-class memory transfer in the stated tests.

Area	Headline result from report	Technical implication
Compute	Geekbench 6: 2017 single-core; 3947 multi-core.	Useful signal for serial response and concurrent processing; validate with application metrics and VM policy.
4K storage	Up to 249k sequential-read IOPS; 94.1k random-read IOPS; p99 values reported for tested 4K patterns.	Supports discussion of transactional and VM storage behavior; validate data protection and application consistency.
64K storage	Up to 6439 MiB/s sequential-read bandwidth and 5919 MiB/s random-read bandwidth.	Supports bulk data movement and analytics discussion; test real transfer paths and overhead.
Network	30.2, 30.7, and 31.0 Gbits/sec with matching sender and receiver rates.	Strong throughput signal; separately validate latency, loss, topology, and application protocol behavior.
Memory	76,597.22 MiB transferred at 7,658.72 MiB/sec; maximum latency 0.45 ms.	Useful signal for memory movement; validate capacity, NUMA, contention, cache, and application behavior.

Bottom line

The benchmark results are appropriate for screening and for choosing workloads for a proof of concept. They should not be read as a universal guarantee for every VM, database, application, or customer topology. The technically defensible decision is to reproduce the tests, run representative workloads, include protection and operational overhead, and agree measurable acceptance criteria before production commitment.

Source note

All headline measurements in this guide are transcribed from the supplied SpeedCloud Supreme Performance Benchmark - June 2026 V5 report. Where the source report does not provide a configuration detail or metric, this guide identifies it as a qualification question rather than filling the gap with an assumption.